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Research Interests

- **Federated and Privacy-Preserving Learning** — training across institutions that cannot share data; non-IID client distributions, aggregation strategies, client drift, and the accuracy cost of decentralisation.
- **Medical Image Analysis** — multi-modal MRI, tumor and lesion segmentation, and the class imbalance that makes small but clinically decisive structures easy to miss.
- **Computer Vision** — semantic segmentation, image classification, and attention mechanisms that earn their parameter budget rather than merely consume it.
- **Lightweight and Efficient Deep Architectures** — parameter-efficient design for deployment on the modest hardware available outside well-funded research labs.
- **Vision-Language and Multimodal Models** — multimodal classification and parameter-efficient fine-tuning under tight memory budgets.
- **Natural Language Processing for Low-Resource Languages** — Bangla, and retrieval-augmented generation grounded in verifiable evidence.

Education

- **Rajshahi University of Engineering and Technology (RUET)** Rajshahi, Bangladesh
  Bachelor of Science in Computer Science and Engineering, CGPA: 3.55 / 4.00 *2021 – 2026*
  - **Thesis**: *Lightweight Double-Attention U-Net for Federated Multi-Class Brain Tumor Segmentation.*
  - **Final four semesters**: 3.73 / 4.00 average.
- **Notre Dame College** Dhaka, Bangladesh
  Higher Secondary Certificate (HSC), Science, GPA: 5.00 / 5.00 *2018 – 2020*
- **Birsreshtha Noor Mohammad Public College** Dhaka, Bangladesh
  Secondary School Certificate (SSC), Science, GPA: 5.00 / 5.00 *2016 – 2018*

Research Experience

- **Undergraduate Thesis Research — *Lightweight Double-Attention U-Net for Federated Multi-Class Brain Tumor Segmentation*** RUET
  Supervisor: A. F. M. Minhazur Rahman, Assistant Professor, Department of CSE, RUET *2025 – 2026*
  - **Problem**: Annotated brain MRI is private patient data, so hospitals will not pool it and each site trains on the small archive it holds. Existing federated segmentation work largely relies on heavy models that ignore the network and compute budgets of the hospitals it is meant to serve.
  - **Architecture**: Designed Lightweight *DA-MDU-Net*, a double-attention multiscale dense U-Net — dense encoder blocks, spatial and channel attention over multi-scale bottleneck features, attention gates on every skip connection, and depthwise separable convolutions throughout. Sharing one depthwise multi-scale extractor across the query, key and value branches cut both attention modules over twenty-fold, taking the model from 97.98 M to **8.19 M parameters (91.6% fewer)** and its arithmetic down 4.49 times.
  - **Federated system**: Built the full training stack — client-side local training, sample-weighted FedAvg aggregation, global validation, checkpointing with automatic resume and per-round metric logging — over 802 adult glioma cases from BraTS 2023 partitioned by whole patient across three clients, for twenty rounds of two local epochs. Group normalization throughout, so no running statistics are averaged across clients, and a loss combining Dice, focal, boundary and region terms to protect the smallest sub-regions.
  - **Results**: **0.8819 mean region Dice** on 80 held-out cases (whole tumor 0.9023, tumor core 0.9033, enhancing tumor 0.8402), scoring every slice including those with no tumor. Two ablations isolate the costs: against centralized training at a matched budget federation costs **0.55 points** of mean region Dice, and against the 97.98 M full-capacity re-implementation the lightweight redesign costs **nothing detectable** (0.8787 vs. 0.8819) while cutting a full run's traffic from 43.80 GB to 3.66 GB.

- **External validation**: Run without fine-tuning on 94 BraTS-Africa cases — a Sub-Saharan African cohort on older, lower-field scanners — it holds 0.7533 mean region Dice (median case 0.8081), with precision high and sensitivity lower: it misses tumor rather than inventing it, the failure a radiologist can correct.
- **Stated limits**: one seed per arm, an IID rather than site-based partition, a 2D formulation, and communication cost computed rather than measured — motivating next steps in non-IID partitions, differential privacy and 3D.

- **DA-MDU-Net — Centralized Implementation and Public Release** 2025 – 2026
  Independent research project — the architecture the thesis federates
  - Implemented and trained the DA-MDU-Net architecture centrally on BraTS 2020 as a 2D pipeline over the T1ce, T2 and FLAIR modalities, segmenting the necrotic core, peritumoral edema and enhancing tumor. Publicly released as the reference implementation of the architecture. [GitHub]
- **Parameter-Efficient Vision-Language Fine-Tuning for a Low-Resource Language** 2026
  Independent research project
  - Fine-tuned Qwen2-VL-2B-Instruct with 4-bit QLoRA on a single 16 GB Kaggle T4 GPU to classify Bangla memes into four categories; prepared, labelled and split the dataset, and selected the two-epoch checkpoint through per-class error analysis rather than the best validation score, reaching 82.07% accuracy and 83.24% macro accuracy on a 290-image held-out split. [GitHub]
- **Machine Learning for Dengue Diagnosis from Routine Hematological Data** 2026
  Independent research project
  - Built the preprocessing and feature pipeline on a 1,523-patient clinical blood dataset (outlier removal, SMOTE, consensus feature selection across five methods) and compared models from logistic regression through gradient-boosted trees under stratified cross-validation; LightGBM led at 76.57% accuracy and 0.71 ROC-AUC, with platelet and lymphocyte counts the strongest predictors. [GitHub]

Manuscripts in Preparation

The following manuscript is in preparation and is the intended first submission from my thesis work.

- **Eazdan Mostafa Rafin** and A. F. M. Minhazur Rahman. “Lightweight Double-Attention U-Net for Federated Multi-Class Brain Tumor Segmentation.” *Manuscript in preparation*, 2026. Drawn from the completed undergraduate thesis: an 8.19 M-parameter federated segmentation model reaching 0.8819 mean region Dice on BraTS 2023, with cross-cohort validation on BraTS-Africa. Target: a peer-reviewed venue in medical image analysis or applied deep learning.

Professional Experience

- **Industrial Trainee** March 2025
  Vivasoft Limited *Rajshahi, Bangladesh*
  - **Contributions**: Designed and built *CV Insighter*, a FastAPI and PostgreSQL system that ingests unstructured PDF and DOCX CVs, scores each against a job description with LangChain and the Groq API, and returns a ranked shortlist. [GitHub]
  - **Working Practice**: Worked inside a professional engineering team under a formal code-review process, with requirements agreed across roles and every change reviewed before it was accepted.

Technical Projects

Artificial Intelligence and Machine Learning

- **RAG Document Question-Answering API**: Built an ingestion pipeline that chunks PDF and DOCX with LangChain, embeds with all-MiniLM-L6-v2 and indexes into FAISS for top-k retrieval, served behind FastAPI with chunk-level citations back to the source page; added a two-tier prompt-injection guard (regex pre-screen in front of an LLM classifier) benchmarked for precision, recall, F1 and latency on a hand-labelled set. [GitHub]

Backend Systems and APIs

- **CERA: Content Signal Extraction and Recommendation API**: Built a five-service backend (authentication, YouTube, Reddit, LLM, gateway) with JWT and Redis-OTP authentication and per-service Alembic migrations; scored public engagement signals with a transparent weighted formula to rank trending content, and cut upstream latency with three-tier caching (Redis, PostgreSQL, external API) and `asyncio.gather` fan-out. [GitHub]

- **Expense Tracker API**: A deployed FastAPI and PostgreSQL backend with authentication, per-user expense CRUD and optional LLM-assisted entry and insights, with a separate front end. [GitHub], [Live]

Honors, Awards and Competitions

- **Kaggle Playground Series S5E7, Predict the Introverts from the Extroverts**: ranked 525 of 4,329 teams (top 13%) in an open international competition, as a team. [Kaggle]
- **DL Sprint 4.0 Datathon, BUET CSE Fest 2026**: ranked 51 of 124 teams on a Bengali long-form speech recognition problem, as a team. [Kaggle]
- **HackSpark AI-API Hackathon, Technocracy Lite 2026, RUET ECE**: finished in the top 10, scoping, building and presenting a working solution as a team inside the contest window. [Certificate]

Certifications and Training

- **Supervised Machine Learning: Regression and Classification** — DeepLearning.AI and Stanford University (Coursera), July 2024. [Verify]
- **Python for Data Analysis: Pandas and NumPy** — Coursera Project Network, July 2024. [Verify]

Technical Skills

- **Programming Languages**: Python, C, C++, SQL, JavaScript
- **Machine Learning and Deep Learning Tools**: PyTorch, scikit-learn, Hugging Face Transformers, LightGBM, XGBoost, CatBoost, NumPy, Pandas, Matplotlib, OpenCV
- **Deep Learning Methods**: CNNs, U-Net / DenseNet, attention mechanisms, semantic and medical image segmentation, image classification, vision-language models, federated learning, PEFT / LoRA / QLoRA, Unsloth
- **LLM and Retrieval**: LangChain, FAISS, Sentence-Transformers, Gemini and Groq APIs, Ollama
- **Backend and Databases**: FastAPI, REST APIs, SQLAlchemy, Alembic, PostgreSQL, MySQL, MongoDB, Redis
- **Tools**: Git, Docker, Docker Compose, Linux, LaTeX, Jupyter, Google Colab and Kaggle GPU
- **Languages**: English (speaking — fluent, writing — fluent), Bangla (native)

Relevant Coursework

- **Artificial Intelligence and Data**: Machine Learning, Artificial Intelligence, Data Mining, Digital Image Processing, Statistics and Probability.
- **Core Computer Science**: Data Structures and Algorithms, Algorithm Design and Analysis, Database Systems, Operating Systems, Computer Networks.
- **Software and Systems**: Object-Oriented Programming, Software Engineering.

Teaching Interests

- **Courses prepared to teach**: Programming Fundamentals (C / C++ / Python), Data Structures and Algorithms, Algorithm Design and Analysis, Database Systems, Artificial Intelligence, Machine Learning, Digital Image Processing, Software Engineering, and the associated laboratory courses.
- **Supervision interests**: undergraduate project and thesis work in medical image analysis, efficient deep learning architectures, and applied machine learning on clinical and low-resource-language data.

References

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